Work Allocation Portal

IT Requirements Document

ISM Data Technology | 16 October 2026 | Prepared for: Muthu Kulathur, Founder & CEO

1. Overview

ISM needs one web portal where admin allocates work files to vendors, vendors download only their own files within a deadline, and QC and payment status stay visible to everyone concerned.

Purpose: Replace manual file sharing (mail, WhatsApp, drive links) with a controlled, trackable allocation system.

Scope (Phase 1):

- Admin allocates files/batches to vendors with a deadline
- Vendor logs in, views allocation, downloads assigned files
- Vendor raises a work request (how many files needed); request also goes to admin email
- QC status and payment status visible per file/batch

User roles:

Role	Who	Main purpose
Admin	ISM management / allocation team	Allocate work, approve requests, set deadlines, update payments
QC	ISM QC team	Review submitted files, mark Approved / Rework / Rejected
Vendor	Sub-vendors and partner teams	Download files, request work, upload output, check status

2. Work Allocation and Download

Admin allocates numbered files (for example 1, 2, 3, 4, 5) to a vendor, sets a deadline such as 3 days, and the vendor downloads only those files from the portal.

- **Allocation:** Admin selects vendor, selects files or a batch range, sets deadline (days or exact date/time), adds optional instructions.
- **Vendor view:** Vendor sees only files allocated to them: file ID, name, allocated date, deadline, countdown, status.
- **Download:** Single-file download and bulk ZIP download. Every download is logged (who, when, IP).
- **Deadline rules:** Timer starts at allocation. Reminder at 24 hours and 4 hours before. Overdue files turn red and alert admin.
- **Reallocation:** Admin can extend deadline, reassign a file to another vendor, or take it back (with reason).
- **Submission:** Vendor uploads completed output against each file; status moves to Submitted.
- **Access control:** A vendor can never see or download another vendor's files.

3. Work Request Module

A vendor first requests the number of files they can handle; the request reaches ISM by email and in the admin dashboard, and files are allocated only after admin approval.

1. Vendor opens **New Work Request** and enters: number of files needed, project/type, preferred deadline, remarks.
2. Portal saves the request (status: Pending) and sends an email to official@ismdatatech.com with vendor name, count and a link to approve.
3. Admin reviews the request and chooses Approve full, Approve partial (edit count) or Reject with reason.
4. On approval, the portal auto-picks the next available files from the pool (or admin picks manually), sets the deadline and allocates.
5. Vendor gets an email and an in-portal notification that files are ready for download.

Rules:

- Admin can set a maximum open-request limit per vendor (for example no new request while overdue files exist)
- Request history stays visible to both vendor and admin
- Daily cap per vendor is configurable

4. QC Status and Payment Tracking

Vendors see, for every file they finish, its QC result and whether payment is cleared, so they never need to ask admin for status.

File status lifecycle:

Status	Meaning	Set by
Allocated	File assigned, not yet downloaded	System
In Progress	Vendor downloaded / working	System
Submitted	Vendor uploaded output	Vendor
In QC	Waiting for or under QC review	System / QC
Rework	QC found errors; vendor must fix and resubmit	QC
Approved	QC passed	QC
Rejected	Output unusable; no payment	QC
Overdue	Deadline crossed without submission	System

QC details visible to vendor: QC date, error count, error category, QC remarks, rework deadline, rework count.

Payment status:

Payment status	Meaning
Not Eligible	File not yet Approved
Pending	Approved, payment not yet processed
Processing	Included in current payment cycle
Paid	Paid, with date and reference number
On Hold	Held due to dispute or pending rework

Vendor dashboard widgets: Files allocated today, due soon, overdue, in QC, rework pending, approved, amount payable, amount paid. Rate per file and total payable are calculated automatically from approved files only.

5. Technical Requirements

A role-based web application with a relational database, secure file storage and email notifications.

Core data entities:

Entity	Key fields
User	id, name, email, role, vendor_id, status
Vendor	id, name, contact, rate per file, payment terms
File	id, name, project, storage path, size, status
Allocation	id, file_id, vendor_id, allocated_at, deadline, extended_deadline
WorkRequest	id, vendor_id, files_requested, files_approved, status, remarks
Submission	id, allocation_id, output path, submitted_at, version
QCRecord	id, submission_id, qc_by, result, error_count, remarks
Payment	id, vendor_id, period, amount, status, paid_on, reference
AuditLog	id, user_id, action, entity, timestamp, IP

Screens:

- Admin: Dashboard, Vendors, File Pool, Allocate Work, Requests, QC Overview, Payments, Reports, Settings
- QC: My Queue, Review Screen, Rework Tracker
- Vendor: Dashboard, My Files, New Request, Upload Output, QC Status, Payments

Notifications (email + in-portal): New request to admin; allocation done; deadline reminder; overdue alert; QC result; rework raised; payment released.

Reports: Vendor-wise productivity, on-time delivery %, QC error rate, rework rate, payment summary, all exportable to Excel/CSV.

Non-functional requirements:

- Login with password policy and optional OTP; session timeout
- Role-based access; vendor sees only own data
- Encrypted file storage (for example AWS S3) with expiring download links
- Audit log for every download, upload, status change
- Daily backup; 99% uptime; supports 100+ concurrent users
- Responsive, works on mobile browser
- Files up to 500 MB and ZIP bulk download

6. Workflow, Phases and Open Questions

Every file reaches payment only through QC approval.

Workflow:

1. Vendor requests N files
2. Request mailed to admin and shown on dashboard
3. Admin approves? (No: rejected with reason, vendor notified)
4. Files allocated with deadline; vendor notified by email
5. Vendor downloads assigned files and works
6. Vendor uploads output
7. QC result? Errors found: Rework raised, vendor fixes and resubmits (loop back to upload)
8. QC pass: file Approved
9. Payment moves from Pending to Paid

Suggested phases:

Phase	Scope
1	Login, roles, file pool, allocation with deadline, vendor download, audit log
2	Work request with admin email, approval, upload output, deadline reminders
3	QC module, rework cycle, payment tracking, vendor dashboard
4	Reports, Excel export, bulk allocation rules, mobile polish

Open questions for ISM:

- Will files be stored on AWS S3 or ISM's own server?
- Should a vendor see rate per file, or only the total payable?
- Auto-allocation from the pool, or always manual admin selection?
- Do we need client-wise or project-wise separation inside the portal?
- Payment cycle: weekly, fortnightly or monthly?